

There may be a bias-variance trade-off between DID_M and the two-way fixed effects regression estimators. For instance, assume that Regression 1 is correctly specified:

$$\begin{aligned} Y_{g,t}(0) &= \alpha_g + \lambda_t + \varepsilon_{g,t}, \\ Y_{g,t}(1) - Y_{g,t}(0) &= \delta, \\ E(\varepsilon_{g,t} | \mathbf{D}) &= 0. \end{aligned}$$

Then, if the errors $\varepsilon_{g,t}$ are homoskedastic and uncorrelated, it follows from the Gauss-Markov theorem that $\hat{\beta}_{fe}$ is the linear estimator of δ , the constant treatment effect parameter, with the lowest variance. As DID_M is also an unbiased linear estimator of δ , the variance of $\hat{\beta}_{fe}$ must be lower than that of DID_M . With heteroskedastic or correlated errors, one can construct examples where the variance of $\hat{\beta}_{fe}$ is higher than that of DID_M , but this still suggests that DID_M may often have a larger variance than that of $\hat{\beta}_{fe}$, as we find in our applications in Section 6.

DID_M uses groups whose treatment is stable to infer the trends that would have affected switchers if their treatment had not changed. This strategy could fail, if switchers experience different trends than groups whose treatment is stable. To assess if this is a serious concern, we propose to use the following placebo estimator, that essentially compares the outcome's evolution from $t - 2$ to $t - 1$, in groups that switch and do not switch treatment between $t - 1$ and t . This placebo estimator is defined under a modified version of Assumption 11.

Assumption 13 (*Existence of “stable” groups for the placebo test*) For all $t \in \{3, \dots, T\}$:

1. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-2} = D_{g,t-1} = 0$ and $D_{g,t} = 1$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-2} = D_{g',t-1} = D_{g',t} = 0$.
2. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-2} = D_{g,t-1} = 1, D_{g,t} = 0$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-2} = D_{g',t-1} = D_{g',t} = 1$.

For all $t \in \{2, \dots, T\}$ and for all $(d, d', d'') \in \{0, 1\}^3$, let

$$N_{d,d',d'',t} = \sum_{g: D_{g,t}=d, D_{g,t-1}=d', D_{g,t-2}=d''} N_{g,t}$$

denote the number of observations with treatment status d'' at period $t - 2$, d' at period $t - 1$, and d at period t . Let

$$\begin{aligned} N_S^{\text{pl}} &= \sum_{(g,t): t \geq 3, D_{g,t} \neq D_{g,t-1} = D_{g,t-2}} N_{g,t}, \\ \text{DID}_{+,t}^{\text{pl}} &= \sum_{g: D_{g,t}=1, D_{g,t-1}=D_{g,t-2}=0} \frac{N_{g,t}}{N_{1,0,0,t}} (Y_{g,t-1} - Y_{g,t-2}) - \sum_{g: D_{g,t}=D_{g,t-1}=D_{g,t-2}=0} \frac{N_{g,t}}{N_{0,0,0,t}} (Y_{g,t-1} - Y_{g,t-2}), \\ \text{DID}_{-,t}^{\text{pl}} &= \sum_{g: D_{g,t}=D_{g,t-1}=D_{g,t-2}=1} \frac{N_{g,t}}{N_{1,1,1,t}} (Y_{g,t-1} - Y_{g,t-2}) - \sum_{g: D_{g,t}=0, D_{g,t-1}=D_{g,t-2}=1} \frac{N_{g,t}}{N_{0,1,1,t}} (Y_{g,t-1} - Y_{g,t-2}). \end{aligned}$$

When there is no group such that $D_{g,t} = 1, D_{g,t-1} = D_{g,t-2} = 0$ or no group such that $D_{g,t} = D_{g,t-1} = D_{g,t-2} = 0$, we let $\text{DID}_{+,t}^{\text{pl}} = 0$, and we adopt the same convention for $\text{DID}_{-,t}^{\text{pl}} = 0$. Let

$$\text{DID}_M^{\text{pl}} = \sum_{t=3}^T \left(\frac{N_{1,0,0,t}}{N_S^{\text{pl}}} \text{DID}_{+,t}^{\text{pl}} + \frac{N_{0,1,1,t}}{N_S^{\text{pl}}} \text{DID}_{-,t}^{\text{pl}} \right).$$

Theorem 4 *If Assumptions 1, 2, 4, 5, 9, 10, 12, and 13 hold, then $E \left[\text{DID}_M^{\text{pl}} \right] = 0$.*

$\text{DID}_{+,t}^{\text{pl}}$ compares the evolution of the mean outcome from $t-2$ to $t-1$ in two sets of groups: those untreated at $t-2$ and $t-1$ but treated at t , and those untreated at $t-2, t-1$, and t . If Assumptions 4 and 5 hold, then $E \left[\text{DID}_{+,t}^{\text{pl}} \right] = 0$. Similarly, if Assumptions 9 and 10 hold, $E \left[\text{DID}_{-,t}^{\text{pl}} \right] = 0$. Then, $E \left[\text{DID}_M^{\text{pl}} \right] = 0$ is a testable implication of Assumptions 4, 5, 9, and 10, so finding DID_M^{pl} significantly different from 0 would imply that those assumptions are violated: groups that switch treatment experience different trends before that switch than the groups used to reconstruct their counterfactual trends when they switch.¹⁵ Note that DID_M^{pl} compares the trends of switching and stable groups one period before the switch. One can define other placebo estimators comparing those trends, say, two or three periods before the switch. DID_M^{pl} and all those other placebo estimators are computed by the `did_multiplegt` Stata package.

5 Extensions

In this section, we briefly review some of the extensions in our Web Appendix. First, we show that the decomposition of β_{fe} in Theorem 1 can be extended to fuzzy designs where the treatment varies within (g, t) cells, to applications with a non binary treatment, and to two-way fixed effects regressions with control variables.¹⁶ In fuzzy designs or with a non-binary treatment, the weights in Theorem 1 remain essentially unchanged.

We also consider two-way fixed effects regressions with covariates. Specifically, we study the coefficient of $D_{g,t}$ in a regression of $Y_{i,g,t}$ on group and period fixed effects, $D_{g,t}$, and a vector of covariates $X_{g,t}$. We show that a result very similar to Theorem 1 applies to that coefficient, up to two differences. First, including covariates allows for different trends across groups, provided those differential trends are fully accounted for by a linear model in $X_{g,t} - X_{g,t-1}$, the change in a group's covariates. Specifically, instead of Assumptions 4 and 5, one needs to assume that

$$E(Y_{g,t}(0) | \mathbf{D}_g, \mathbf{X}_g) - E(Y_{g,t-1}(0) | \mathbf{D}_g, \mathbf{X}_g) = (X_{g,t} - X_{g,t-1})' \gamma + \lambda_t,$$

¹⁵See also Callaway and Sant'Anna (2018), who propose another placebo test in staggered adoption designs.

¹⁶The decomposition of β_{fd} in Theorem 2 can also be extended to all of those cases.

for some vector γ and constant λ_t , and where $\mathbf{X}_g = (X_{g,1}, \dots, X_{g,T})$. Importantly, when the covariates are group-specific linear trends, the equation above is equivalent to

$$E(Y_{g,t}(0)|\mathbf{D}_g, \mathbf{X}_g) - E(Y_{g,t-1}(0)|\mathbf{D}_g, \mathbf{X}_g) = \gamma_g + \lambda_t,$$

meaning that from $t - 1$ to t , the evolution of $Y(0)$ in group g should deviate from its group-specific linear trend γ_g by an amount λ_t common to all groups. Second, the residual $\varepsilon_{g,t}$ in the weights in Theorem 1 has to be replaced by $\varepsilon_{g,t}^X$, the residual of observations in cell (g, t) in the regression of $D_{g,t}$ on group and period fixed effects and $X_{g,t}$. Some of the corresponding weights may still be negative, as in Theorem 1. Overall, two-way fixed effects regressions with covariates may rely on a more plausible common trends assumptions than those without covariates, but they still require that the treatment effect be homogeneous, across time and between groups.

Third, we show that under the common trends assumption and the assumption that the ATE of a (g, t) cell does not change over time, β_{fe} and β_{fd} identify weighted sums of the ATEs of the (g, t) cells whose treatment changes between $t - 1$ and t . In sharp designs, the weights attached to β_{fd} are all positive, while for β_{fe} , the same only holds in staggered adoption designs.

Fourth, we show that our DID_M estimator can easily be extended to non-binary, discrete treatments. Then, we define it as a weighted average of DIDs comparing the evolution of the outcome in groups whose treatment went from d to d' between $t - 1$ and t and in groups with a treatment of d at both dates, across all possible values of d , d' , and t .

Finally, our `twowayfeweights`, `fuzzydid`, and `did_multipligt` Stata packages can handle all of those extensions.

6 Applicability, and applications

6.1 Applicability

We conducted a review of all papers published in the American Economic Review (AER) between 2010 and 2012 to assess the importance of two-way fixed effects regressions in economics. Over these three years, the AER published 337 papers. Out of these 337 papers, 33 or 9.8% of them estimate the FE or FD Regression, or other regressions resembling closely those regressions. When one withdraws from the denominator theory papers and lab experiments, the proportion of papers using these regressions raises to 19.1%.

Table 1: Papers using two-way fixed effects regressions published in the AER

	2010	2011	2012	Total
Papers using two-way fixed effects regressions	5	14	14	33
% of published papers	5.2%	12.2%	11.2%	9.8%
% of empirical papers, excluding lab experiments	12.8%	23.0%	19.2%	19.1%

Notes. This table reports the number of papers using two-way fixed effects regressions published in the AER from 2010 to 2012.

Table 2 shows descriptive statistics about the 33 2010-2012 AER papers estimating two-way fixed effects regressions. Panel A shows that 13 use the FE regression; six use the FD regression; six use regressions the FE or FD regression with several treatment variables; three use the FE or FD 2SLS regression discussed in Section ?? of the Web Appendix; five use other regressions that we deemed sufficiently close to the FE or FD regression to include them in our count.¹⁷ Panel B shows that more than three fourths of those papers consider sharp designs, while less than one fourth consider fuzzy designs. Finally, Panel C assesses whether, in those applications, there are groups whose exposure to the treatment remains stable between each pair of consecutive time periods, the condition that has to be met to be able to compute the DID_M estimator. For about a half of the papers, reading the paper was not enough to assess this with certainty. We then assessed whether they presumably have stable groups or not. Overall, 12 papers have stable groups, 14 presumably have stable groups, five presumably do not have stable groups, and two do not have stable groups.

In Section ?? of the Web Appendix, we review each of the 33 papers. We explain where two-way fixed effects regressions are used in the paper, and we detail our assessment of whether the design is a sharp or a fuzzy design, and of whether the stable groups assumption holds or not.

¹⁷For instance, two papers use regressions with three-way fixed-effects instead of two-way fixed effects.